NAGA ARJUN MARICHAMY

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ASPIRING VLSI ENGINEER | RTL DESIGN,VERIFICATION & PD

PROFILE SUMMARY

An ambitious and results-driven Electronics & Communication Engineering student with a solid foundation in **VLSI Design, Digital Systems, and Verification**. Skilled in **Verilog, RTL design, testbench basics**, and familiar with industry-standard EDA tools like **Vivado and EDA Playground.** Strong problem-solving ability with hands-on projects including ALU, FSM controllers, and embedded prototypes.

TECHNICAL SKILLS

- **HDLs:** Verilog
- **EDA Tools:** Xilinx Vivado, EDA Playground,GTK wave
- **Digital Design:** RTL Design, Digital Logic, FSM Design, ASIC/FPGA Design Flow
- **Programming Languages:** C, Python
- **Operating Systems:** Linux (Ubuntu), Windows
- **Version Control:** Git, GitHub

PROJECT EXPERIENCE

1. 4-bit ALU (Verilog RTL + Testbench)

- Designed a 4-bit ALU supporting ADD, SUB, AND, OR, XOR operations.
- Fully coded in Verilog and simulated using Vivado/EDA Playground.
- Verified using waveform analysis.

2. Traffic Light Controller (Moore FSM)

- Implemented a 3-state Traffic Light Controller using Moore FSM architecture.
- Designed clean state transitions and simulated using testbench.
- Validated waveform for RED → GREEN → YELLOW cycle.

EDUCATION

- Government College of Engineering, Erode, Tamil Nadu
- B.E. in Electronics & Communication Engineering - **CGPA** (**7.85/10)** | 2023- 2027
- Government Higher Secondary School,Ramanathapuram,Tamil Nadu
- **HSC** - **87.6%** | 2021 - 2023

COURSES & CERTIFICATIONS

- **Python – GUVI**🔗 -May,2024
- **VLSI for Beginners –** NIELIT -March,2025
- **Digital System Design –** NPTEL -May,2025
- **Fundamentals of Digital Design for VLSI Chip Design –** L&T EduTech🔗 -Sep,2025
- **VLSI Design Flow from RTL to GDS Workshop –** ChipXpert Technologies
- **Digital Design with Verilog –** NPTEL -May,2026

AWARDS & ACHIEVEMENTS

- Runner-up at InnovateX Hackathon (2025), won Rs. 15000 for a V2V prototype
- Best Innovation Award, HackTronix Hardware Hackathon (2025), won Rs. 2000
- Ranked Top 70 among 2500+ in SPRINTATHON'25 for Carmunicate (V2V system)